Vishalpreet Singh

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Summary

Aspiring Data Scientist with a strong foundation in Machine Learning and Agentic AI. Proficient in Python and C++, with a focus on developing intelligent, data-driven solutions and automating complex workflows.

Technical Skills

- **AI & Data Science:** Machine Learning, Agentic AI, Predictive Modeling, NumPy, Pandas, Scikit-Learn.
- **Languages:** Python, C++, SQL, Java, JavaScript.
- **Databases:** PostgreSQL, MongoDB, MySQL.
- **Server-Side & Architecture:** Node.js, Express.js, RESTful APIs, JWT Authentication.
- **Client-Side Development:** React.js, TypeScript, Next.js, Tailwind CSS.
- **Cloud & Tools:** AWS (EC2/S3), Docker, Git/GitHub.

Relevant Coursework

Data Science	Machine Learning	Algorithms Analysis	Data Structures
Database Management	Software Methodology	React JS	Node JS

Projects

Github - Retrieval-Augmented Generation (RAG) for QnA System - End-to-End Generative AI Implementation

Tech Stack - Python, LangChain, OpenAI/Ollama, ChromaDB, Ragas, Gradio, PyPDF, Hugging Face Transformers, Linux/Bash

- Engineered an end-to-end RAG pipeline using a "Load-Chunk-Embed-Retrieve" architecture to enable precise, context-aware querying over unstructured PDF datasets.
- Implemented advanced retrieval strategies, including recursive character text splitting and Hybrid Search (Vector + Keyword), to significantly reduce model hallucinations and improve context recall.
- Optimized system performance by integrating a Re-ranking step (Cross-Encoders) to prioritize the most relevant document chunks before generation, ensuring high-fidelity responses.
- Systematically evaluated model performance using the Ragas framework, specifically tracking the "RAG Triad" (Faithfulness, Answer Relevance, and Context Precision) to guide iterative improvements.
- Developed an interactive playground using Gradio for real-time model testing and visualization of retrieved context, bridging the gap between back-end AI logic and a user-friendly presentation layer

Github-Campus Connect - Campus Connect - Frontend & Backend Development

Tech Stack - React JS, Type Script, JavaScript, Node JS, PHP, Express JS, Vite, Tailwind CSS, Mongo DB.

- Build a user friendly interface for seamless usage across various decisions that integrates academics and facilitated seamless access to academic resources, to empower students in their learning journey.
- Enabled real-time collaboration & notifications via WebSockets / Firebase Cloud Messaging.
- Archived PYQs for competitive and academic exams with subject-wise and topic-wise labels.
- Backend structured to be adaptable for PHP/Laravel-based API migration, offered mock tests and quizzes with instant evaluation and feedback.
- Used visual analytics using AI for analysis of progress and performance of users.

Certificates

Bluestock Fintech - Software Development Engineer Intern, Bluestock Fintech (Dec 2025 – Jan 2026): Developed and optimized fintech solutions using industry-standard engineering practices, focusing on scalable architecture and robust system logic

Infosys SpringBoard - Completed structured training in Python and C, focusing on problem-solving, algorithms, and programming fundamentals.

Think NEXT Technologies - Completed hands-on training in ReactJS, AWS cloud services, and full-stack web development concepts

Dataiku Certified Professional: Completed a comprehensive certification path including ML Practitioner, Advanced Designer, and Core Designer, covering the full machine learning lifecycle from data orchestration to model deployment.

Education

Guru Nanak Dev Engineering College, Ludhiana | Bachelor's Degree in Computer Science & Engineering | 2022-26

Holy Family Convent School, SHGP Gurdaspur | ISC 10+2 PCM | 2021-22 | CISCE

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Coding Profile

Hacker Rank - 3-star rating in DSA (Java)

Geeks for Geeks - 100+ problems solved, focusing on DP and graph theory.